

The Large-Divisor Prime-Pair Defect at Almost All Shifts: Uniform Small-Divisor Transfer, Fourier Coordinates, and Finite-Conductor Energy Escape

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July 2026

Abstract

For a cutoff D , write the von Mangoldt function as the exact sum of its short source-divisor part and its complementary defect,

$$a_D(n) = - \sum_{\substack{d|n \\ d \leq D}} \mu(d) \log d, \quad r_D(n) = \Lambda(n) - a_D(n).$$

The preceding paper in this series isolated

$$\mathcal{T}_{h,>D}(X) = \sum_{X < n \leq 2X} r_D(n) \Lambda(n+h)$$

as the complete large-divisor remainder left after a Bombieri–Vinogradov evaluation of the prime-pair correlation. For a fixed nonzero even shift, estimating this remainder by $o(X)$ is equivalent to the expected Hardy–Littlewood asymptotic and is therefore open.

Here the quantifier on the shift is changed and nothing else is removed from the defect. We first prove, uniformly for $1 \leq |h| \leq 2X^{1-\varepsilon}$, that the short-divisor correlation with $D = X^{1/2}(\log X)^{-B}$ equals $\mathfrak{S}(h)X + O_A(X(\log X)^{-A})$. Combining this uniform transfer with the averaged Hardy–Littlewood theorem of Matomäki, Radziwiłł, and Tao shows that, whenever

$$X^{8/33+\varepsilon} \leq H \leq X^{1-\varepsilon},$$

the full defect satisfies

$$\mathcal{T}_{h,>D}(X) = O_{A,\varepsilon}(X(\log X)^{-A})$$

for all but $O_{A,\varepsilon}(H(\log X)^{-A})$ shifts in every permitted window of length $2H$. The deep almost-all prime-pair theorem is an external input; the contribution here is the uniform defect transfer and its exact interpretation. Fixed h , in particular $h = 2$, is not controlled.

We place the result in additive Fourier coordinates: the defect correlations are matrix coefficients of the integer translation representation and their global square sum is an exact weighted Fourier energy. This is a spectral identity, not a new dispersion estimate. Finally, if V_q denotes the space of q -periodic functions on $(N, 2N]$, we prove uniformly for $q \leq N^{1-\delta}$ that

$$\|P_{V_q}(\Lambda - 1)\|_2^2 \ll_\delta N \frac{q}{\varphi(q)} \ll_\delta N \log \log(3q),$$

whereas $\|\Lambda - 1\|_2^2 \sim N \log N$. Thus sublinear finite-conductor models capture a vanishing proportion of the prime fluctuation's linear L^2 energy. This is a periodic-approximation boundary, not an impossibility theorem for nonlocal sieve or Type II methods. No fixed-shift prime-pair lower bound is proved.

Keywords: prime-pair defect; von Mangoldt correlation; almost-all shifts; Bombieri–Vinogradov theorem; Fourier translation; finite-conductor approximation.

MSC 2020: Primary 11N36; Secondary 11N13, 11N35, 42A38, 47B15.

1 Introduction and strict scope

For a nonzero integer h and $X \geq 2$, put

$$\mathcal{C}_h(X) = \sum_{X < n \leq 2X} \Lambda(n) \Lambda(n+h), \quad (1)$$

with the convention $\Lambda(t) = 0$ for $t \leq 0$. The Hardy–Littlewood prediction is

$$\mathcal{C}_h(X) \sim \mathfrak{S}(h)X \quad (2)$$

for each fixed even h , where

$$\mathfrak{S}(h) = \begin{cases} 2 \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{\substack{p|h \\ p>2}} \frac{p-1}{p-2}, & 2 \mid h, \\ 0, & 2 \nmid h. \end{cases} \quad (3)$$

Even an $o(X)$ error in (2) is not known for any single nonzero even shift; the case $h = 2$ would imply the twin-prime conjecture. This paper does not alter that fixed-shift fact.

The elementary identity

$$\Lambda(n) = - \sum_{d|n} \mu(d) \log d \quad (4)$$

gives, for every $1 \leq D \leq 2X$, the exact decomposition

$$a_D(n) = - \sum_{\substack{d|n \\ d \leq D}} \mu(d) \log d, \quad (5)$$

$$r_D(n) = \Lambda(n) - a_D(n) = - \sum_{\substack{d|n \\ d > D}} \mu(d) \log d, \quad (6)$$

and hence

$$\mathcal{C}_h(X) = \mathcal{C}_{h, \leq D}(X) + \mathcal{T}_{h, > D}(X), \quad (7)$$

where

$$\mathcal{C}_{h, \leq D}(X) = \sum_{X < n \leq 2X} a_D(n) \Lambda(n+h), \quad (8)$$

$$\mathcal{T}_{h, > D}(X) = \sum_{X < n \leq 2X} r_D(n) \Lambda(n+h). \quad (9)$$

The terminology *source divisor* refers to a divisor introduced by expanding the first von Mangoldt factor in (4).

TPC-6 proved for each fixed nonzero even h that classical Bombieri–Vinogradov theory evaluates (8) at $D = X^{1/2}(\log X)^{-B}$, and consequently that (2) is equivalent to $\mathcal{T}_{h, > D}(X) = o(X)$ [11]. That equivalence named the missing object but supplied no estimate for it. The present paper retains the same complete remainder and changes the topology in the shift variable: pointwise control is replaced by control outside a set of asymptotic density zero in each sufficiently long shift window.

1.1 Main contributions and provenance

The logical inputs and outputs are separated as follows.

- R1.** We prove a singular-series coefficient estimate uniform for $1 \leq |h| \leq z^K$. Together with the maximal Bombieri–Vinogradov theorem, it yields

$$\mathcal{C}_{h, \leq D}(X) = \mathfrak{S}(h)X + O_A(X(\log X)^{-A})$$

simultaneously for all $1 \leq |h| \leq 2X^{1-\varepsilon}$. The analytic ingredients are classical; the point is the uniformity needed for the growing shift window.

- R2.** We use the averaged Hardy–Littlewood theorem of Matomäki et al. [6, Theorem 1.3(i)]. It gives the same main term for the full correlation outside $O_{A,\varepsilon}(H(\log X)^{-A})$ shifts when $H \geq X^{8/33+\varepsilon}$. Subtraction proves that the *same full defect* from TPC-6 is small at almost all such shifts. The exponent $8/33$ and the deep prime-correlation estimate are entirely due to those authors; we do not claim a new circle-method, dispersion, or Type II estimate.
- R3.** We record the exact Fourier representation of the defect and its Plancherel energy. This gives a rigorous translation-dynamical interface, but the identity alone contains no cancellation estimate.
- R4.** We prove a finite-conductor energy-escape theorem. Periodic functions of every modulus $q \leq N^{1-\delta}$ capture only $O_\delta(\log \log(3q)/\log N)$ of the L^2 energy of $\Lambda - 1$. This quantifies a linear periodic-approximation boundary and does not extend to arbitrary nonlinear or nonlocal information.

Earlier almost-all-shift prime-pair results culminated, for the present purpose, in the theorem of Matomäki, Radziwiłł, and Tao, which improved the shift-window threshold from $X^{1/3+\varepsilon}$ to $X^{8/33+\varepsilon}$ [6, 7]. Our result is not a further improvement of that threshold. It transfers their theorem to the exact large-source-divisor coordinate singled out by the preceding operator and sieve analysis.

1.2 What is not claimed

For clarity, none of the following is asserted.

- B1.** The exceptional set is not shown to avoid any prescribed shift. In particular, it may contain $h = 2$ for every X .
- B2.** No individual dyadic layer of (6) is shown to be small. Cancellation may occur between different divisor scales.
- B3.** The Fourier identity is not a spectral gap, an operator isomorphism, or an estimate for a fixed Fourier coefficient.
- B4.** The finite-conductor theorem concerns orthogonal approximation by a single periodic linear space. It is not a universal parity-barrier theorem and does not rule out growing bilinear, dispersion, automorphic, or other nonlocal methods.
- B5.** No lower bound for twin primes, no fixed-shift asymptotic, and no new level of distribution for the primes is proved.

2 A uniform small-divisor transfer theorem

The fixed-shift coefficient calculation in TPC-6 is not by itself sufficient for a window in which h grows with X : its displayed error constant was allowed to depend on h . We first remove that dependence over every polynomial range. Uniform truncated singular-series estimates of this type also occur in the divisor-sum correlation work of Goldston and Yıldırım [2, Lemma 2.1]; a coefficient-level proof adapted to the present weight is included for completeness.

2.1 The singular-series coefficient uniformly in the shift

Lemma 2.1 (Uniform singular-series coefficient). *Fix $A, K > 0$. Uniformly for $z \geq 3$ and integers h satisfying $1 \leq |h| \leq z^K$,*

$$- \sum_{\substack{d \leq z \\ (d, h) = 1}} \frac{\mu(d) \log d}{\varphi(d)} = \mathfrak{S}(h) + O_{A, K}((\log z)^{-A}). \quad (10)$$

Proof. Set

$$a_h(d) = \mathbf{1}_{(d, h) = 1} \frac{\mu(d)}{\varphi(d)}$$

and, initially for $\Re s > 0$, write

$$F_h(s) = \sum_{d \geq 1} \frac{a_h(d)}{d^s} = \frac{G_h(s)}{\zeta(1+s)}. \quad (11)$$

The Euler factors of G_h are

$$G_{h,p}(s) = \begin{cases} \frac{1 - p^{-s}/(p-1)}{1 - p^{-1-s}}, & p \nmid h, \\ (1 - p^{-1-s})^{-1}, & p \mid h. \end{cases} \quad (12)$$

Let g_h be the multiplicative function whose Dirichlet series is G_h . Expanding (12) coefficient by coefficient gives, for $\nu \geq 1$,

$$g_h(p^\nu) = \begin{cases} p^{-\nu}, & p \mid h, \\ -\frac{1}{p^\nu(p-1)}, & p \nmid h. \end{cases} \quad (13)$$

Thus, in the half-plane of absolute convergence and hence as a coefficient identity,

$$a_h(n) = \sum_{rk=n} g_h(r) \frac{\mu(k)}{k}. \quad (14)$$

We need two uniform bounds for g_h . The absolute Euler product at $s = 0$ and its first logarithmic derivative give

$$\sum_{r \geq 1} |g_h(r)| \ll \log \log(3|h|), \quad (15)$$

$$\sum_{r \geq 1} |g_h(r)|(1 + \log r) \ll (1 + \log \log(3|h|))^2. \quad (16)$$

Indeed, the factors with $p \nmid h$ form an absolutely convergent product, while those with $p \mid h$ contribute at most $\prod_{p|h} (1 - p^{-1})^{-1} \ll \log \log(3|h|)$ and $\sum_{p|h} \log p / (p-1) \ll 1 + \log \log(3|h|)$.

For the tail we use a fixed Rankin exponent. If $0 < \delta < 1/2$, then

$$\sum_{r \geq 1} |g_h(r)| r^\delta \ll_\delta \exp\left(C_\delta (1 + (\log(3|h|))^\delta)\right). \quad (17)$$

For $p \nmid h$ the relevant local product converges because its nontrivial part is $O_\delta(p^{-2+\delta})$. For $p \mid h$ it is $(1 - p^{-1+\delta})^{-1}$. The standard prime-rearrangement bound $\sum_{p|h} p^{-1+\delta} \ll_\delta 1 + (\log(3|h|))^\delta$ proves (17). We emphasize that no assertion about arbitrary high logarithmic moments is needed here.

Put

$$U(y) = \sum_{k \leq y} \frac{\mu(k)}{k}, \quad V(y) = \sum_{k \leq y} \frac{\mu(k) \log k}{k}.$$

The classical zero-free-region form of the prime number theorem, followed by partial summation, gives

$$U(y) \ll e^{-c\sqrt{\log y}}, \quad V(y) = -1 + O(e^{-c\sqrt{\log y}}) \quad (y \geq 3), \quad (18)$$

with a possible change in the positive absolute constant c . Using (14), the left side of (10) equals

$$\sum_{r \leq z} g_h(r) \{-\log r U(z/r) - V(z/r)\}. \quad (19)$$

Split at $r = z^{1/2}$. On the lower range, (18) and (16) replace the quantity in braces by 1 with error $O_{A,K}((\log z)^{-A})$. On the upper range, the elementary bound

$$|\log r U(z/r) + V(z/r)| \ll (\log(2z))^2$$

is sufficient. From (17), the assumptions $|h| \leq z^K$ and $0 < \delta < 1/2$ give

$$\sum_{r > z^{1/2}} |g_h(r)| \ll_{\delta,K} z^{-\delta/2} \exp(O_{\delta,K}((\log z)^\delta)) \ll_{\delta,K} z^{-\delta/3}$$

for sufficiently large z . The same estimate replaces $\sum_{r \leq z^{1/2}} g_h(r)$ by $G_h(0)$. Consequently (19) is $G_h(0) + O_{A,K}((\log z)^{-A})$.

Finally, evaluation of (12) at zero gives

$$G_h(0) = \prod_{p|h} \frac{p}{p-1} \prod_{p \nmid h} \left(1 - \frac{1}{(p-1)^2}\right) = \mathfrak{S}(h). \quad (20)$$

If h is odd, the second product has a zero factor at $p = 2$; if h is even, separating the factor at 2 gives (3). This proves the lemma. $\square$

Remark 2.2 (Why the uniform proof is separate). The fixed- h estimate with an $O_{A,h}$ constant does not imply lemma 2.1 when h grows. The Rankin tail in the proof is what absorbs the mild growth of the Euler factors belonging to primes dividing h .

2.2 Uniform evaluation of the short source-divisor part

For real $y \geq 0$ and a reduced residue class $a \pmod{q}$, put

$$E(y; q, a) = \sum_{\substack{n \leq y \\ n \equiv a \pmod{q}}} \Lambda(n) - \frac{y}{\varphi(q)}. \quad (21)$$

Lemma 2.3 (Uniform non-coprime contribution). *Fix $0 < \varepsilon < 1/2$. Uniformly for $1 \leq |h| \leq 2X^{1-\varepsilon}$ and $1 \leq D \leq 2X$,*

$$\sum_{\substack{d \leq D \\ (d,h) > 1}} |\mu(d)| \log(2d) \sum_{X < dm \leq 2X} \Lambda(dm + h) \ll_{\varepsilon, \eta} X^\eta \quad (22)$$

for every $\eta > 0$, provided X is sufficiently large.

Proof. Write $n = dm$. If $(d, h) > 1$, choose a prime $p \mid (d, h)$. Then $p \mid n + h$. The condition $\Lambda(n + h) \neq 0$ forces $n + h = p^j$. For sufficiently large X , the target interval lies in $(X/2, 3X]$. Consequently only $O_\varepsilon(1)$ powers p^j lie in $(X + h, 2X + h]$ for each $p \mid h$. For each resulting n ,

$$\sum_{d|n} |\mu(d)| \log(2d) \ll \tau(n) \log(2n) \ll_\eta X^\eta.$$

The number of prime divisors of h and the von Mangoldt weight cost only powers of $\log X$, which are absorbed after decreasing η . $\square$

Theorem 2.4 (Uniform sharp small-divisor transfer). *Fix $A > 0$ and $0 < \varepsilon < 1/2$. There is $B = B(A, \varepsilon) > 0$ such that, with*

$$D = \left\lfloor \frac{X^{1/2}}{(\log X)^B} \right\rfloor, \quad (23)$$

one has

$$\mathcal{C}_{h, \leq D}(X) = \mathfrak{S}(h)X + O_{A, \varepsilon} \left(\frac{X}{(\log X)^A} \right) \quad (24)$$

uniformly for all integers

$$1 \leq |h| \leq 2X^{1-\varepsilon}. \quad (25)$$

Proof. Expanding a_D and writing $n = dm$ gives

$$\mathcal{C}_{h, \leq D}(X) = - \sum_{d \leq D} \mu(d) \log d \sum_{X < dm \leq 2X} \Lambda(dm + h). \quad (26)$$

The terms with $(d, h) > 1$ contribute $O_{A, \varepsilon}(X(\log X)^{-A})$ by [lemma 2.3](#), after choosing a fixed small $\eta < 1$.

If $(d, h) = 1$, the inner sum is

$$\frac{X}{\varphi(d)} + E(2X + h; d, h) - E(X + h; d, h). \quad (27)$$

The maximal Bombieri–Vinogradov theorem states, after taking B large enough in terms of the requested logarithmic saving, that

$$\sum_{d \leq D} \max_{(a, d)=1} \max_{y \leq 3X} |E(y; d, a)| \ll_{A, \varepsilon} \frac{X}{(\log X)^{A+2}}. \quad (28)$$

The range (25) ensures that both endpoints in (27) lie below $3X$ for sufficiently large X . Multiplication by $\log d$ costs at most one logarithm, so the total progression error in (26) is $O_{A, \varepsilon}(X(\log X)^{-A-1})$.

The continuous main term is exactly

$$-X \sum_{\substack{d \leq D \\ (d, h)=1}} \frac{\mu(d) \log d}{\varphi(d)}.$$

Because $D = X^{1/2+o(1)}$ and $|h| \leq 2X^{1-\varepsilon} \leq D^3$ for all sufficiently large X , [lemma 2.1](#) with $K = 3$ evaluates it as $\mathfrak{S}(h)X + O_{A, \varepsilon}(X(\log X)^{-A})$. Combining the three parts proves the theorem. $\square$

Remark 2.5 (Classical analytic input). The estimate (28) is the classical maximal Bombieri–Vinogradov theorem [[1](#), [3](#), [10](#)]. [Theorem 2.4](#) is a uniform application of that theorem, not a new distribution result for primes.

3 Almost-all-shift closure of the complete defect

We now change only the quantifier on h . The deep input is quoted in the exact sharp-interval form needed here.

Theorem 3.1 (Matomäki–Radziwiłł–Tao). *Let $A > 0$, $0 < \varepsilon < 1/2$, and*

$$\sigma = \frac{8}{33}. \quad (29)$$

Suppose

$$X^{\sigma+\varepsilon} \leq H \leq X^{1-\varepsilon}, \quad 0 \leq h_0 \leq X^{1-\varepsilon}. \quad (30)$$

Then

$$\mathcal{C}_h(X) = \mathfrak{S}(h)X + O_{A,\varepsilon} \left(\frac{X}{(\log X)^A} \right) \quad (31)$$

for all but

$$O_{A,\varepsilon} \left(\frac{H}{(\log X)^A} \right) \quad (32)$$

integers h satisfying $|h - h_0| \leq H$.

This is Matomäki et al. [6, Theorem 1.3(i)]. Their convention includes $\Lambda(n) = 0$ for nonpositive n . We exclude $h = 0$ below, since the diagonal correlation has a different scale. One excluded coordinate is absorbed by (32) in the range (30).

Theorem 3.2 (Complete-defect closure at almost all shifts). *Fix $A > 0$ and $0 < \varepsilon < 1/2$. There exists $B = B(A, \varepsilon) > 0$ such that, for*

$$D = \left\lfloor \frac{X^{1/2}}{(\log X)^B} \right\rfloor, \quad (33)$$

and every (H, h_0) in the range (30), one has

$$\mathcal{T}_{h, > D}(X) = O_{A,\varepsilon} \left(\frac{X}{(\log X)^A} \right) \quad (34)$$

for all but

$$O_{A,\varepsilon} \left(\frac{H}{(\log X)^A} \right) \quad (35)$$

nonzero integers h with $|h - h_0| \leq H$.

Proof. Every shift in the stated window satisfies

$$|h| \leq h_0 + H \leq 2X^{1-\varepsilon}.$$

Choose the logarithmic saving in theorem 2.4 slightly larger than A . Uniformly for every nonzero h in the window, it gives

$$\mathcal{C}_{h, \leq D}(X) = \mathfrak{S}(h)X + O_{A,\varepsilon} \left(\frac{X}{(\log X)^A} \right). \quad (36)$$

Outside the exceptional set in theorem 3.1, the full correlation satisfies the same formula. Subtract (36) from (31) and use the exact identity (7). $\square$

Remark 3.3 (What is genuinely controlled). Theorem 3.2 estimates the entire sum over all source divisors $d > D$. It is not a surrogate tail, a truncated Type II packet, or a statement in which an extra coefficient has been removed. On the other hand, the proof does not estimate any prescribed divisor layer separately; different layers may cancel in the complete sum.

Remark 3.4 (Provenance and priority). The prime-pair asymptotic and the exponent $8/33$ in theorem 3.1 are not new. Earlier work treated longer shift windows, and Matomäki et al. [6] obtained the quoted threshold. Large-/small-divisor decompositions in parity problems also have important precedents, including Murty and Vatswani [9]. The theorem here is a transfer statement for the exact TPC-6 defect coordinate, supported by the uniform small-divisor theorem of section 2; it is not presented as a new almost-all prime-pair theorem.

3.1 Density and fixed- L^p formulations

Let

$$\mathcal{H}(X; H, h_0) = \{h \in \mathbb{Z} \setminus \{0\} : |h - h_0| \leq H\}. \quad (37)$$

Corollary 3.5 (Shift-density convergence). *Fix $A > 0$ and $0 < \varepsilon < 1/2$. There is $B = B(A, \varepsilon) > 0$ such that, with D as in (33), for every (H, h_0) in (30),*

$$\frac{1}{H} \# \left\{ h \in \mathcal{H}(X; H, h_0) : \frac{|\mathcal{T}_{h, > D}(X)|}{X} > (\log X)^{-A} \right\} \ll_{A, \varepsilon} (\log X)^{-A}. \quad (38)$$

Proof. Apply [theorem 3.2](#) with requested saving $A + 2$. For sufficiently large X , its implicit constant times $(\log X)^{-A-2}$ is at most $(\log X)^{-A}$. Every shift above the displayed threshold therefore belongs to that theorem's exceptional set, whose relative size is $O_{A, \varepsilon}((\log X)^{-A-2})$. This proves the stated, weaker bound. $\square$

Corollary 3.6 (Fixed- L^p shift closure). *Fix $p \geq 1$, $A > 0$, and $0 < \varepsilon < 1/2$. There is $B = B(A, p, \varepsilon) > 0$ such that, with D as in (33), for every (H, h_0) in (30),*

$$\left(\frac{1}{\#\mathcal{H}(X; H, h_0)} \sum_{h \in \mathcal{H}(X; H, h_0)} \left| \frac{\mathcal{T}_{h, > D}(X)}{X} \right|^p \right)^{1/p} \ll_{A, p, \varepsilon} (\log X)^{-A}. \quad (39)$$

Proof. The elementary pointwise bounds

$$\begin{aligned} |\mathcal{C}_h(X)| &\ll X(\log(3X))^2, \\ |\mathcal{C}_{h, \leq D}(X)| &\leq \log(3X) \sum_{d \leq D} \log(2d) \left(\frac{X}{d} + 1 \right) \ll X(\log(3X))^3 \end{aligned}$$

give

$$|\mathcal{T}_{h, > D}(X)| \ll X(\log(3X))^3. \quad (40)$$

Choose a fixed auxiliary exponent $R > p(A + 3) + 1$ and apply [theorem 3.2](#) with requested saving R . On the regular shifts use that theorem, and on its $O(H(\log X)^{-R})$ exceptional set use (40). Since

$$\frac{1}{\#\mathcal{H}(X; H, h_0)} \sum_{h \in \mathcal{H}(X; H, h_0)} \left| \frac{\mathcal{T}_{h, > D}(X)}{X} \right|^p \ll_{R, p, \varepsilon} (\log X)^{-Rp} + (\log X)^{3p-R},$$

taking the p th root and absorbing the fixed constants proves (39). $\square$

The freedom to request arbitrary logarithmic saving in [theorem 3.1](#) is what allows every fixed p . No claim is made for p growing with X or for the supremum norm.

4 Translation dynamics and additive spectral coordinates

The shift variable has an exact and elementary dynamical realization. This section records it in order to distinguish the resulting identity from the analytic estimate imported in [theorem 3.1](#).

Put $e(x) = \exp(2\pi i x)$. For a finitely supported sequence $f : \mathbb{Z} \rightarrow \mathbb{C}$, define

$$\hat{f}(\theta) = \sum_{n \in \mathbb{Z}} f(n) e(n\theta), \quad \theta \in \mathbb{T} = \mathbb{R}/\mathbb{Z}. \quad (41)$$

Proposition 4.1 (Exact Fourier representation of correlations). *Let $f, g : \mathbb{Z} \rightarrow \mathbb{R}$ be finitely supported and set*

$$C_h(f, g) = \sum_{n \in \mathbb{Z}} f(n) g(n + h). \quad (42)$$

Then

$$C_h(f, g) = \int_{\mathbb{T}} \overline{\widehat{f}(\theta)} \widehat{g}(\theta) e(-h\theta) d\theta, \quad (43)$$

$$\sum_{h \in \mathbb{Z}} |C_h(f, g)|^2 = \int_{\mathbb{T}} |\widehat{f}(\theta)|^2 |\widehat{g}(\theta)|^2 d\theta. \quad (44)$$

Equivalently, if $U^h g(n) = g(n + h)$ is the unitary integer-translation representation on $\ell^2(\mathbb{Z})$, then

$$C_h(f, g) = \langle f, U^h g \rangle_{\ell^2(\mathbb{Z})}. \quad (45)$$

Proof. Expand the integrand in (43). Fourier orthogonality retains exactly the pairs $t = n + h$. Formula (44) is Parseval applied to the Fourier coefficients in (43), and (45) is the definition of U^h . $\square$

For the defect application, let $Y = 2X^{1-\varepsilon}$ and define

$$f_{X,D}(n) = r_D(n) \mathbf{1}_{X < n \leq 2X}, \quad (46)$$

$$g_{X,Y}(t) = \Lambda(t) \mathbf{1}_{X-Y < t \leq 2X+Y}. \quad (47)$$

For every $|h| \leq Y$,

$$\mathcal{T}_{h,>D}(X) = C_h(f_{X,D}, g_{X,Y}) = \langle f_{X,D}, U^h g_{X,Y} \rangle. \quad (48)$$

Thus the complete defect is an inverse Fourier coefficient of

$$\overline{\widehat{f}_{X,D}(\theta)} \widehat{g}_{X,Y}(\theta). \quad (49)$$

The global energy identity is

$$\sum_{h \in \mathbb{Z}} |C_h(f_{X,D}, g_{X,Y})|^2 = \int_{\mathbb{T}} |\widehat{f}_{X,D}(\theta)|^2 |\widehat{g}_{X,Y}(\theta)|^2 d\theta. \quad (50)$$

Remark 4.2 (Identity versus estimate). Neither (49) nor (50) shows that a coefficient is small. The analytic content in [theorem 3.2](#) comes from Bombieri–Vinogradov for the short divisor coordinate and from the theorem of Matomäki, Radziwiłł, and Tao for the full correlation. Parseval does not replace those inputs.

Remark 4.3 (A triangular family, not a fixed mixing system). Both vectors in (48), the cutoff D , and the window grow with X . Therefore [corollary 3.6](#) is a statement about a triangular family of translation matrix coefficients. It is not a mixing theorem for one fixed pair of vectors, a spectral gap for U , or a spectral isomorphism with a logistic map.

4.1 Why density convergence does not select a fixed shift

The topology in [corollary 3.6](#) deliberately forgets a set of vanishing relative size. Evaluation at one coordinate is not uniformly continuous in that topology.

Proposition 4.4 (Fixed-coordinate spike). *Let $\mathcal{H}_X \subset \mathbb{Z}$ be finite sets with $|\mathcal{H}_X| \rightarrow \infty$, all containing a fixed integer h_* . Define*

$$s_X(h) = \mathbf{1}_{h=h_*}. \quad (51)$$

Then, for every fixed $1 \leq p < \infty$,

$$\left(\frac{1}{|\mathcal{H}_X|} \sum_{h \in \mathcal{H}_X} |s_X(h)|^p \right)^{1/p} = |\mathcal{H}_X|^{-1/p} \rightarrow 0, \quad (52)$$

while $s_X(h_*) = 1$ for every X .

Consequently, no abstract passage from (39) to a bound at $h = 2$ is possible. In fact the allowance $O(H(\log X)^{-A})$ in (35) grows without bound in the permitted power-sized windows, so the exceptional set may contain any preassigned finite collection of shifts at every scale.

5 Finite-conductor energy escape and a layerwise boundary

The almost-all-shift theorem uses genuinely nonlocal information about prime correlations. We next quantify why simply fitting the prime fluctuation by a periodic observable does not recover comparable L^2 energy.

5.1 Orthogonal projection onto periodic observables

Let

$$I_N = \{n \in \mathbb{Z} : N < n \leq 2N\} \quad (53)$$

and equip $\ell^2(I_N)$ with counting measure. Let $V_q(N)$ be the subspace of restrictions to I_N of q -periodic complex functions, and let P_q be the orthogonal projection onto $V_q(N)$.

Theorem 5.1 (Finite-conductor energy escape). *Fix $0 < \delta < 1$. Uniformly for integers*

$$1 \leq q \leq N^{1-\delta}, \quad (54)$$

one has

$$\|P_q(\Lambda - 1)\|_{\ell^2(I_N)}^2 \ll_\delta N \frac{q}{\varphi(q)} \ll_\delta N \log \log(3q). \quad (55)$$

On the other hand,

$$\|\Lambda - 1\|_{\ell^2(I_N)}^2 \sim N \log N. \quad (56)$$

Consequently, uniformly in (54),

$$\frac{\|P_q(\Lambda - 1)\|_2^2}{\|\Lambda - 1\|_2^2} \ll_\delta \frac{\log \log(3q)}{\log N} = o_\delta(1), \quad (57)$$

$$\inf_{b \in V_q(N)} \|\Lambda - 1 - b\|_2^2 = (1 - o_\delta(1))N \log N. \quad (58)$$

Proof. For $a \pmod{q}$, let

$$N_a = \#\{n \in I_N : n \equiv a \pmod{q}\}. \quad (59)$$

The range (54) gives $N_a = N/q + O(1) \asymp_\delta N/q$ uniformly for sufficiently large N . For any $f \in \ell^2(I_N)$, the projection is the residue-class average, so

$$\|P_q f\|_2^2 = \sum_{a \pmod{q}} \frac{1}{N_a} \left| \sum_{\substack{n \in I_N \\ n \equiv a \pmod{q}}} f(n) \right|^2. \quad (60)$$

Let $\Lambda_{\text{pr}}(n) = \log n$ when n is prime and zero otherwise. If $(a, q) = 1$, the Brun–Titchmarsh inequality and $\log(N/q) \geq \delta \log N$ imply

$$\sum_{\substack{N < p \leq 2N \\ p \equiv a \pmod{q}}} \log p \ll_\delta \frac{N}{\varphi(q)}. \quad (61)$$

We use here only the classical Brun–Titchmarsh inequality; the other standard prime-number estimates in this proof may be found, for example, in Montgomery and Vaughan [8]. Thus the

residue sum of $\Lambda_{\text{pr}} - 1$ has absolute value $O_\delta(N/\varphi(q))$. If $(a, q) > 1$, no prime $p > N > q$ lies in that class, and the residue sum is simply $-N_a$. Substitution into (60) yields

$$\|P_q(\Lambda_{\text{pr}} - 1)\|_2^2 \ll_\delta N \frac{q}{\varphi(q)} + N \ll_\delta N \frac{q}{\varphi(q)}. \quad (62)$$

The prime-power remainder $R_{\text{pp}} = \Lambda - \Lambda_{\text{pr}}$ satisfies

$$\|R_{\text{pp}}\|_2^2 \leq \sum_{\substack{N < p^k \leq 2N \\ k \geq 2}} (\log p)^2 \ll N^{1/2} (\log N)^2. \quad (63)$$

Orthogonal projection is a contraction. Combining (62) and (63), using $\|u+v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$, proves the first bound in (55). The standard estimate $q/\varphi(q) \ll \log \log(3q)$ proves the second.

The prime number theorem and partial summation give

$$\sum_{n \in I_N} \Lambda(n) \sim N, \quad \sum_{n \in I_N} \Lambda(n)^2 \sim N \log N,$$

which proves (56). Finally, the orthogonal decomposition

$$\inf_{b \in V_q(N)} \|\Lambda - 1 - b\|_2^2 = \|(I - P_q)(\Lambda - 1)\|_2^2 = \|\Lambda - 1\|_2^2 - \|P_q(\Lambda - 1)\|_2^2$$

gives (57) and (58). $\square$

Remark 5.2 (Exact boundary of the theorem). The space $V_q(N)$ is a linear finite-conductor model. A small projection onto $V_q(N)$ says nothing by itself about bilinear forms, adaptive weights, growing collections of moduli, or nonlinear sieve information. Thus [theorem 5.1](#) is not a general impossibility theorem for analytic number theory.

5.2 The outer divisor slice already sees shifted-prime Möbius

The exact defect can also be written

$$\mathcal{T}_{h, > D}(X) = - \sum_{D < d \leq 2X} \mu(d) \log d \sum_{X < dm \leq 2X} \Lambda(dm + h). \quad (64)$$

If $d > X$, the inner variable is forced to be $m = 1$. Therefore the outermost slice is exactly

$$\mathcal{O}_h(X) = - \sum_{X < d \leq 2X} \mu(d) \log d \Lambda(d + h). \quad (65)$$

For fixed h , removing target prime powers of exponent at least two gives

$$\mathcal{O}_h(X) = - \sum_{X+h < p \leq 2X+h} \mu(p-h) \log(p-h) \log p + O_h(X^{1/2} (\log X)^2). \quad (66)$$

The neighboring unweighted assertion

$$\sum_{p \leq X} \mu(p+h) = o(\pi(X)) \quad (67)$$

for a fixed nonzero shift is a folklore open problem. Averaged-shift forms of (67) are known [4, 5]. Formula (66) explains why a proposed proof that bounds every divisor slice separately already meets a difficult fixed-shift correlation at the outer edge.

Remark 5.3 (No inference for an individual layer). The smallness of the complete sum in [theorem 3.2](#) does not imply that (65) is small. Other ranges of d in (64) may cancel it. The observation above is a warning against layerwise absolute-value arguments, not a deduction from the complete-tail theorem.

6 Finite diagnostics and conclusion

The theorems above are analytic and do not depend on computation. We nevertheless supply a deterministic finite diagnostic for three purposes: to test the exact divisor split, to expose the slow finite-scale behavior of the tail, and to illustrate the periodic-energy quantity in [theorem 5.1](#).

6.1 FFT defect certificate

The script `experiments/defect_shift_diagnostics.py` constructs μ and Λ by sieving. For each cutoff it forms a_D by adding $-\mu(d) \log d$ on the multiples of d , puts $r_D = \Lambda - a_D$, and computes all correlations

$$\sum_{X < n \leq 2X} r_D(n) \Lambda(n+h), \quad 1 \leq h \leq H,$$

with one zero-padded FFT convolution. Direct dot products at selected shifts and a second FFT for the short part verify

$$\mathcal{C}_h(X) - \mathcal{C}_{h, \leq D}(X) - \mathcal{T}_{h, > D}(X) = 0 \quad (68)$$

up to floating-point roundoff.

The committed run uses $X = 2^{18}$, $H = 2^{15}$, and the exploratory cutoffs $D = \text{round}(X^\theta)$. The even-shift summaries are shown in [table 1](#).

Table 1: Finite even-shift defect diagnostics. Each statistic is normalized by X ; it is not an asymptotic estimate.

θ	D	median $ \mathcal{T} /X$	90% quantile	RMS
0.25	23	0.3053	1.8429	0.9964
0.35	79	0.2801	0.9830	0.5807
0.45	274	0.1284	0.4629	0.3033

The maximum discrepancy in (68) over the three runs is below 5.8×10^{-10} . The sizable normalized tails in [table 1](#) are a useful warning: the theorem contains a non-explicit logarithmic cutoff constant, and small power cutoffs at $X = 2^{18}$ do not numerically realize its asymptotic regime.

6.2 Periodic projection diagnostic

For every residue class modulo q , the script averages $\Lambda - 1$ on that class and evaluates the exact projection formula (60). Selected captured-energy ratios appear in [table 2](#).

Table 2: Finite ratios $\|P_q(\Lambda - 1)\|_2^2 / \|\Lambda - 1\|_2^2$ on $(N, 2N]$.

N	$q = 2$	$q = 30$	$q = 210$
2^{15}	0.1022	0.2811	0.3462
2^{16}	0.0957	0.2632	0.3237
2^{17}	0.0895	0.2462	0.3026
2^{18}	0.0844	0.2322	0.2852

The decrease is qualitatively consistent with (57), whose convergence is only logarithmic. No statistical fit or extrapolation is used in a proof.

The full machine-readable record is `experiments/data/defect-shift-certificate.json`. Its canonical payload SHA-256 is

6F822312CE0699722A0830F98E91189823F3AB9ABE1F47810340E081AFFDAB30.

Six unit tests check the arithmetic arrays, the complete-cutoff identity, the FFT convention, Euler’s totient, the exact periodic projection, and small-input/generator handling.

Remark 6.1 (Computational boundary). The finite values do not verify the hypothesis of an unproved conjecture, choose the exceptional set in [theorem 3.2](#), or provide evidence that $h = 2$ is regular. Their role is reproducibility and finite algebraic validation only.

6.3 Conclusion and the remaining route

TPC-6 isolated the full large-source-divisor remainder and proved that its cancellation at a fixed nonzero even shift is exactly the missing prime-pair content. The present paper replaces the fixed-shift quantifier in that ledger by an almost-all-shift quantifier:

$$\begin{aligned} & \text{uniform Bombieri–Vinogradov} \\ & \text{short-divisor transfer} \quad + \quad \text{MRT almost-all prime-pair theorem} \\ & \implies \text{complete large-divisor defect is small at almost all shifts.} \end{aligned} \tag{69}$$

This gives an explicit estimate for that object within the TPC-6 ledger, but only in shift density. The fixed-coordinate spike in [proposition 4.4](#) proves the quantifier gap. The periodic energy escape in [theorem 5.1](#) and the shifted-prime Möbius slice (66) mark boundaries for two natural strategies that might otherwise be proposed to specialize (69) to a prescribed even shift.

One plausible route toward a subsequent fixed-shift advance would seek a coefficient-stable nonlocal estimate. A rigorous but narrower intermediate target is a low-conductor second-coefficient transfer, in which a periodic factor observable is passed through the prime target by Bombieri–Vinogradov theory. Such a theorem can describe exact arithmetic involutions at growing polylogarithmic conductor, but [theorem 5.1](#) shows why it must not be confused with approximation of the full weight $\Lambda - 1$. The broad fixed-shift Type II problem remains open.

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